Lab 4, Act 1

If we knew the truth, would our trend analysis find it?

NRES 746

Answer key. Answers are in magenta. Simulation results come from one run; yours will differ slightly.

The scenario

A state wildlife agency is planning a long-term monitoring program for a rare, secretive toad in a remote Great Basin spring complex. The management goal is to detect a population decline of 25% or more over 25 years.

Here is the 'truth' for this made-up scenario, which the analyst of real-world data never sees: each year a survey produces a count index (# individuals captured or observed in each survey) I_t for years $t = 1, 2, \ldots, 25$. The population declines by the same proportion every year, and each year's count differs from the trend by a random multiplicative factor (proportional observation error). This scatter reflects noise in how well the survey captures the truth (weather, effort, or observer conditions can shift a whole year's count up or down together – this is called "observation error"), not the population itself fluctuating (we could model this too- this is called "process error").

Quantity	Value
Expected index in year 0	I_0 = 60
Total change from year 0 to year 25	-25% (falls to 75% of its year-0 value)
Year-to-year scatter (SD of the log of the multiplicative factor)	sigma = 0.2
Surveys	one per year, t = 1, . . . , 25

The analyst (treating this as real-world data and not knowing the 'truth) sees only the 25 observed counts, fits a straight line to the *log* counts using `lm()`, and reports the slope with a 90% confidence interval. **Across many replicate monitoring programs, does that slope hit the truth on average, how variable is it, and does the interval contain the truth 90% of the time?**

Please close your laptop.

Parts 1-3 and the group check-in that follows are pen-and-paper only- but feel free to use a calculator or use your phone as a calculator.

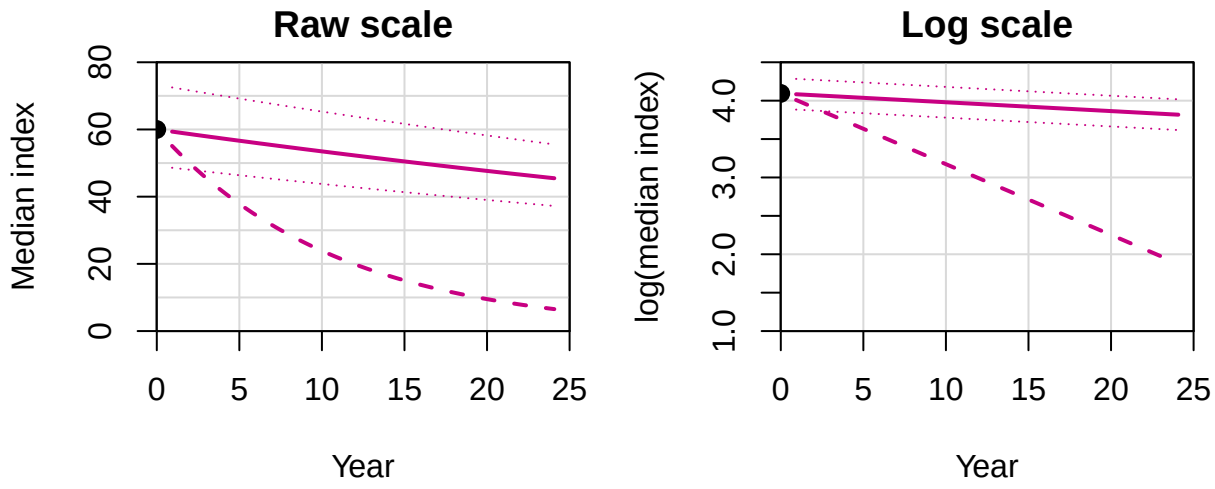
Part 1: The model (individual, ~9 min)

1a. A decline of 25% over 25 years means the index value in year 25 is 0.75 times the index value in year 0. If the index is multiplied by the same factor λ every year, what is λ? Let $b = \ln \lambda$. What is b (also known as intrinsic rate of exponential growth/decline)?

$\lambda^{25} = 0.75$, so $\lambda = 0.75^{1/25}$ and $b = \ln(0.75)/25$.

λ = 0.9886 b = -0.01151 (about 1.15% decline per year)

1b. Sketch the trajectory of the index for a 25% decline and, as a dashed curve, for a 90% decline. Do this on the raw-scale axes at left, then again on the log-scale axes at right. Both start at $I_0 = 60$ in year 0 (the dot).



Which of your curves are straight lines, and why?

Only the log-scale curves (magenta, above): $\log(\text{median}_t) = \log(I_0) + bt$ is linear in t. On the raw scale, $I_0\lambda^t$ is exponential, so both curves bend.

1b, continued. The scatter around the trend has the same spread on the log scale, $\sigma = 0.2$, every single year. On the log-scale axes, add a band of $\pm\sigma$ around your **25% decline** line: a pair of lines running parallel to it, each about 0.2 log-units above and below (constant vertical distance apart the whole way across). Now sketch what that same band looks like (approximately) on the raw-scale axes once it's converted back (roughly: multiply the center point by $e^{0.2} \approx 1.2$ for the top and $e^{-0.2} \approx 0.8$ for the bottom). [note: technically the center point is the median, since the expectation of an exponentiated random variable is actually larger than the exponential of the expectation, by Jensen's inequality - but don't worry if you don't see that just yet]

If we used the raw response (abundance index) in a linear regression, what assumption(s) might we be violating?

Constant variance (homoscedasticity): on the raw scale the band (dotted, left) is about 24 units wide in year 0 and 18 in year 25, a fixed percentage of the median rather than a fixed width. Linearity (the raw trend is curved) and normality of residuals (the noise is lognormal) are also violated.

1c. In 1a you found λ (with $b = \ln\lambda$) so that the population is multiplied by λ every year, with no randomness – that per-year multiplication, applied t times, is the **deterministic decline**: λ^t, equivalently e^{bt}. Combine it with a random multiplicative factor e^{ε_t}, $\varepsilon_t \sim \text{Normal}(0, \sigma)$, independent across years, to write the full model for the index I_t on the raw scale: I_0 times the deterministic decline, times the random factor. Then take the log of both sides to get a model for $\log(I_t)$. Write out this model - does this model meet the basic assumptions of linear regression?

$$I_t = I_0\,\lambda^t\,e^{\varepsilon_t} = I_0\,e^{bt+\varepsilon_t}, \qquad \varepsilon_t \sim \text{Normal}(0,\sigma)$$
$$\log(I_t) = \log(I_0) + bt + \varepsilon_t$$

Yes, on the log scale: linear in t, with additive, independent, normal errors of constant variance σ^2. (The raw-scale version is not.)

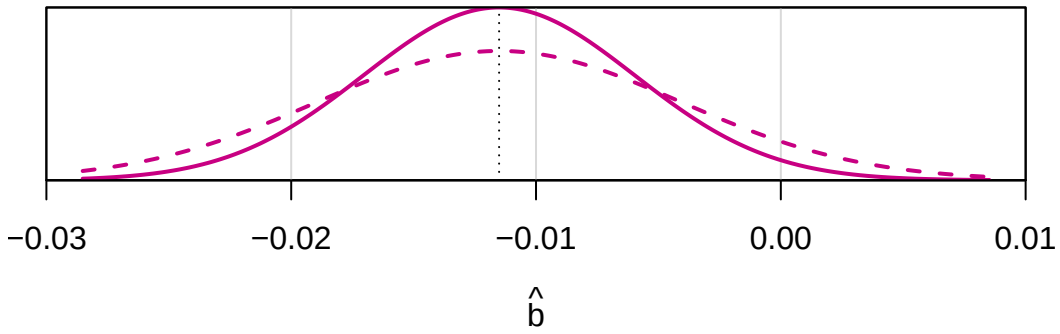
Part 2: What should the simulation show? (individual, ~8 min)

The analyst regresses $z_t = \log(I_t)$ on t and gets a slope estimate $\hat{b}$. Because the counts are random, every simulated monitoring program (replicate) gives a slightly different $\hat{b}$, even though the underlying trend is exactly the same every time – $\hat{b}$ is itself a random variable, with its own distribution across replicate monitoring programs.

2a. Suppose we simulate many replicate monitoring programs under two survey designs: (i) an annual survey ($t = 1, \ldots, 25$, 25 surveys) and (ii) a survey only every other year ($t = 1, 3, \ldots, 25$, 13 surveys). For each design, where would you expect the resulting values of $\hat{b}$, across many replicate monitoring programs, to be centered? Would you expect the two designs to have the same center? Which design do you expect to give a more variable (spread-out) set of $\hat{b}$ values, and how different do you expect the variation (standard error) to be?

Both centered on the true b (the slope estimate is unbiased whichever years are surveyed). Every other year is more variable, since it uses 13 points instead of 25: about 1.34 times the SD of the annual design (see 4C).

2b. On the axes below, sketch what you expect the *distribution* of $\hat{b}$ to look like under each design: a solid curve for annual surveys, a dashed curve for every-other-year surveys. Mark the true value b (from 1a). You don't need exact numbers – just show where each curve is centered and how wide or narrow it is, relative to the other. Remember that the area under each of these curves should be 1.



2c. In a moment you will estimate the coverage of the 90% interval as the fraction of R simulated programs whose interval contains b. Treat each program/replicate as a Bernoulli trial with success probability $p = 0.9$. Using the rules for the variance of a mean (and the fact that the variance of a Bernoulli r.v. is p*(1-p)), what is the standard error of the estimated coverage, as a function of R? How large must R be for that standard error to be less than or equal to 0.01?

SE $= \sqrt{0.9 \times 0.1/R} = \sqrt{0.09/R}$. $\sqrt{0.09/R} \leq 0.01 \Rightarrow R \geq 900$.

Part 3: Plan the experiment (individual, ~3 min)

3. Recall the questions posed at the start: across many replicate monitoring programs, does the slope estimate $\hat{b}$ hit the truth (b) on average, how variable is it, and does the 90% confidence interval contain the truth 90% of the time? Write, in plain English, the step-by-step procedure for a simulation experiment that answers these questions. Do not write R code- instead just write in plain English. State what gets repeated, what you keep track of on each repeat, and what you compute at the end.

Repeat many times (e.g. 2,000): simulate 25 years of counts from the model with the *true* b and σ; fit `lm(log(count) ~ t)`; record $\hat{b}$ and whether its 90% interval contains the true b. At the end: the mean of the $\hat{b}$ values (vs. b, for bias), their SD (variability), and the fraction of intervals containing b (coverage, vs. 0.90). Repeat for the every-other-year design.

Group check-in: finalize your plan, then choose your path (group, ~7 min)

Compare your answers to Parts 2 and 3 with your group. Agree on one version of your speculation about $\hat{b}$'s distribution under each design, your sample-size calculation, and the plain-English procedure. Where you differ, argue it out.

Your group must now pick **one** of the two paths below for running the experiment. Both are equally valid; decide as a group before opening any laptops.

- **Path A (AI-assisted):** write a prompt describing your procedure and the model, and have an AI assistant generate the R code.
- **Path B (self-coded):** write the R code yourselves, using R's documentation and each other, but no AI tools.

Our group is using: Path A (AI-assisted) / Path B (self-coded) *(circle one)*

Laptops open now. Complete only the section below that matches your group's chosen path.

Part 4A: AI-assisted path (laptops, ~20 min)

Use these settings: $I_0 = 60$, $\sigma = 0.2$, b from 1a, and 2,000 simulated programs per design. The analysis is `lm()` of $\log(I_t)$ on t, with a 90% confidence interval for the slope from `confint()`.

Write a prompt describing your group's procedure and asking the AI to write R code that (1) runs the experiment for annual surveys ($t = 1, \ldots, 25$), (2) reports the mean and SD of $\hat{b}$ and the estimated coverage of the 90% interval, (3) draws a histogram of the $\hat{b}$ values with the true b marked and a normal curve (using the simulated mean and SD) overlaid, and (4) repeats the experiment for surveys every other year ($t = 1, 3, \ldots, 25$). **You don't need to submit your code for this path** – just your prompt notes and results below.

Notes on the prompt you used and any changes you had to make to get the AI to do the right thing (record your prompt and what happened, not the AI's code):

Open-ended.

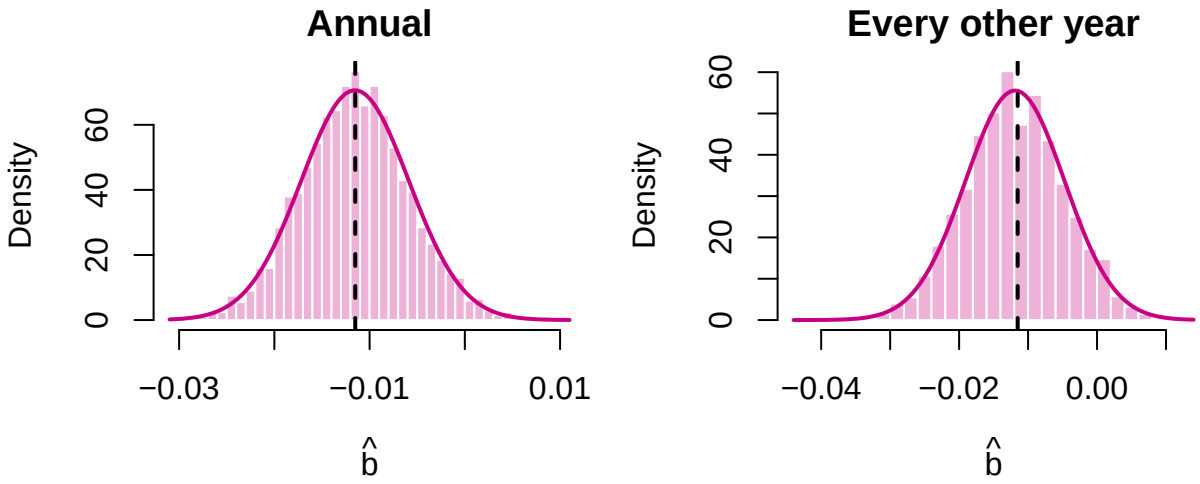
Results, annual surveys. Mean of $\hat{b}$ = -0.01153 SD of $\hat{b}$ = 0.00565

Coverage of the 90% interval = 0.898 ± 0.007 (Monte Carlo standard error)

Results, every other year. Mean of $\hat{b}$ = -0.01189 SD of $\hat{b}$ = 0.00718

How do the simulated results compare with your speculation in 2a and your sketch in 2b?

Both centered on b; every other year about 1.27 times as variable. Your numbers will differ slightly (Monte Carlo); coverage should be near 0.90 for both designs (every other year: 0.905 here).



Reflection. Did the AI's first attempt do what your procedure said? How did your own understanding or lack of understanding of the model from Parts 1-2 affect your ability to get AI to do what you wanted?

Open-ended.

Part 4B: Self-coded path (laptops, ~20 min)

Use these settings: $I_0 = 60$, $\sigma = 0.2$, b from 1a, and 2,000 simulated programs per design. The analysis is `lm()` of $\log(I_t)$ on t, with a 90% confidence interval for the slope from `confint()`.

Write the R code for your group's procedure (R's built-in help, documentation, and human-written websites and neighbors are fair game; AI tools are not). Run the experiment for annual surveys ($t = 1, \ldots, 25$), then for surveys every other year ($t = 1, 3, \ldots, 25$). Draw a histogram of the $\hat{b}$ values for annual surveys with the true b marked. **You don't need to write your code out by hand** – one member should email your group's R script to the instructor, with everyone in your group listed on the email.

Results, annual surveys. Mean of $\hat{b}$ = -0.01153 SD of $\hat{b}$ = 0.00565

Coverage of the 90% interval = 0.898 ± 0.007 (Monte Carlo standard error)

Results, every other year. Mean of $\hat{b}$ = -0.01189 SD of $\hat{b}$ = 0.00718

How do the simulated results compare with your speculation in 2a and your sketch in 2b?

Both centered on b; every other year about 1.27 times as variable (see the histograms under 4A). Your numbers will differ slightly.

Reflection. What part of turning your procedure into working R gave your group the most trouble? How did you resolve it?

Open-ended.

Part 4C: Compare with the formula (both paths, ~5 min)

It turns out there is an exact formula for the sampling distribution of $\hat{b}$:

$$\hat{b} \sim \text{Normal}\left(b,\ \frac{\sigma}{\sqrt{S_{tt}}}\right), \qquad S_{tt} = \sum_t (t - \bar{t})^2.$$

In words: $\hat{b}$ is exactly unbiased (centered on the true b), and its standard deviation shrinks as S_{tt} (how spread out the survey years are) grows. For our two designs, $S_{tt} = 1300$ for annual surveys and $S_{tt} = 728$ for every-other-year surveys.

4C. Using $\sigma = 0.2$, compute the formula's SD of $\hat{b}$ for each design, and compare with your simulated (Monte Carlo) SD from your Results above.

Annual: formula SD = $0.2/\sqrt{1300} \approx 0.00555$ simulated SD = about 0.00565

Every other year: formula SD = $0.2/\sqrt{728} \approx 0.00741$ simulated SD = about 0.00718

Do they agree well? If a formula SD and a simulated SD don't agree closely, is that more likely to be Monte Carlo error (not enough replicates) or a mistake somewhere in the simulation code? How could you tell the difference?

Yes, within a few percent. Monte Carlo error shrinks with more replicates and changes from run to run; a bug gives the same wrong answer every rerun. So rerun with a different seed (or more replicates): a gap that persists is a bug.

Group discussion and revision (~5 min)

Compare your results with a different group (ideally one that took the other path). Where the simulation disagreed with a prediction, find out why before moving on.

Feel free to revise any of your answers above based on the discussion (cross out and rewrite; I like to see your thinking evolve).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion ("muddiest points") or topics you want to review or dust off.

Open-ended.